
PDE Reaction–Diffusion Spheroid Modeling for Probabilistic Chemotherapy Drug Search

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Abstract

Bayesian optimization (BO) over finite combinatorial pools is scored, almost universally, by a single scalar measured in a well-mixed monolayer assay. Savitar, a curve-aware interaction-structured kernel, attains the lowest mean regret on such pools by converting per-constituent response curves into activity-gated embeddings coupled through a shared low-rank tensor factorization. Part I condenses that method and its cross-domain evidence. Part II asks a question the original evaluation cannot: whether the combinations a curve-aware optimizer selects remain the best ones once *space* is modeled. We simulate the selections of sixteen BO methods on the NCI-ALMANAC panel (Holbeck et al., 2017) in a spatially resolved spheroid reaction–diffusion model with oxygen-limited proliferation, hypoxic resistance, and finite drug transport. The well-mixed ODE bridge used previously is exactly rank-preserving (Spearman $\rho = -1.0000$); the spheroid is not ($\rho = -0.56$, with 36.5% of within-cell-line pairs re-ordered). In vitro potency explains 23% of the variance in spheroid outcome; one spatial quantity raises this to 86%. Savitar keeps a large advantage over every structural baseline but places fifth of sixteen on spheroid log-kill AUC, and the deficit is mechanistic: outcome tracks residual hypoxic fraction ($\rho = -0.969$) more tightly than potency ($\rho = -0.767$), and Savitar selects combinations 7.5 percentage points *more* potent in vitro that leave 62% more hypoxic tissue. Combination BO should be scored on spatially resolved, outcome-anchored readouts.

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1. Introduction

Drug-combination screening, phase-specific alloy design, and same-day options strategy search are not usually thought of as the same problem, but each is a search over a finite, enumerable pool under a budget that is a small fraction of pool size, with a long-tailed value distribution in which a tiny minority of candidates are useful, and each is paired with prior information that practitioners routinely measure before any combination is queried. Bayesian optimization (BO) is the standard framework for such budgets, yet off-the-shelf kernels (Matérn on raw descriptors, Hamming on identities, Tanimoto on fingerprints) discard the richest available structure: the *per-constituent response curve* (the dose–response curve, the unary phase-boundary curve, the implied-volatility smile). These curves are informative about how each constituent behaves in combination, cheap to obtain, and shared across every combination containing that constituent.

Part I condenses Savitar (Gonehal, 2026), a structured GP kernel that consumes the curve directly through two priors: an *activity gate* vanishing when a constituent is inactive at its operating point, and a *symmetric CP-tensor parameterization* sharing interaction parameters across all subset orders, reducing the effective parameter count to $O(Dq)$ regardless of arity and letting few observations generalize across unseen combinations. It is a drop-in kernel replacement requiring no new acquisition machinery. This material is published work, included as the foundation the extension builds on.

Part II is new. Every result in Part I, and in the combination-BO literature generally, is scored on a number measured in a well-mixed monolayer: percent growth inhibition at a fixed exposure. That number is what the optimizer maximizes and what benchmark regret is computed from. It is also blind to *where* in a tumor the killing happens. We build a spatially resolved spheroid reaction–diffusion model, re-score the selections of sixteen BO methods under it, and find the method ranking changes, for reasons that are mechanistically identifiable and that suggest a concrete extension to the kernel itself.

2. Related Work

Neural combination surrogates reach strong retrospective accuracy on 10^4 – 10^5 labeled triples, but sequential discovery operates in the opposite regime (≤ 30 evaluations against 10^4 + candidates) where deep surrogates have poorly calibrated uncertainty. Combinatorial and discrete BO innovate on acquisition or search geometry (COMBO’s graph Cartesian product, BOCS’s sparse binary regression, trust-region methods); Savitar attacks the kernel side instead, consuming the per-constituent curve as a structured prior rather than relearning it. Among set-function and tensor-structured kernels, DeepSets (Zaheer et al., 2017) learns set functions without an activity gate or CP coupling and SAAS (Eriksson & Jankowiak, 2021) is a sparsity-aware GP; CP decompositions (Bro, 1997) appear in multi-output GPs and collaborative filtering. Savitar combines CP interaction weights with activity-gated embeddings for finite-pool discovery. Reaction–diffusion models of avascular spheroids with oxygen-limited proliferation are long established as descriptions of viable-rim/hypoxic/ necrotic layering; to our knowledge they have not previously been used as an *evaluation instrument* for combination-selection algorithms, which is the use we make of one here.

3. Method

The pipeline at a glance. The shared input across domains is a finite candidate pool together with the per-constituent response curves that domain practitioners measure before any combination is queried. The paradigm pipeline feeds constituent identities (and, optionally, chemistry-derived fingerprints) to a generic Hamming or Tanimoto kernel, discarding the curve information entirely; the resulting similarity kernel treats every pair of identical constituents the same regardless of where on the dose–response or phase-boundary curve they sit. Savitar instead consumes each curve through an activity-gated linear projection that produces a per-constituent embedding f_i , then aggregates across constituents through a symmetric CP-tensor parameterization that shares interaction parameters A across all subset sizes. The output is a single feature vector $\phi(c) \in \mathbb{R}^q$ on which a Gaussian RBF kernel is computed. The GP posterior on the unobserved pool, scored by Expected Improvement, picks the next query, and the loop repeats until the budget is exhausted.

3.1. Problem formulation

A combinatorial discovery problem is a triple $(\mathcal{C}, f, T)$: a finite candidate pool $\mathcal{C}$ of size P , an oracle $f : \mathcal{C} \rightarrow \mathbb{R}$ returning a scalar quality value when queried, and a budget $T \ll P$ of permitted queries. Each candidate is a tuple $c = (i_1, \dots, i_k; d_1, \dots, d_k)$ of k constituent identities $i_j \in [D]$ and corresponding operating points $d_j \in \mathbb{R}$. The

objective is to find $c^* \in \arg \min_{c \in \mathcal{C}} f(c)$ within T queries; performance is reported as final regret $|f(c_T) - f(c^*)|$. This abstraction covers drugs at log-doses against a cell-line oracle ($D \approx 100$, $k = 2$, $P \approx 4.5 \times 10^4$); elements at mole fractions against a phase-equilibrium oracle; and option contracts at quantities against a one-day P&L oracle. The four properties that recur and motivate Savitar’s design are: pools are finite and enumerable, budgets are small fractions of P , high-value candidates are rare, and each constituent admits an individual-response curve fit upstream of any combination measurement.

3.2. Per-constituent fingerprints

For each constituent i in a given context we fit a fixed parametric model to its individual-response data and store the parameter vector $e_{i,c} \in \mathbb{R}^p$ as a fingerprint. The choice of curve family is a domain question (Hill, polynomial, implied-volatility smile); p is small, typically four or five. The fingerprint admits a deterministic readout $\text{curve}(d; e_i) \rightarrow \mathbb{R}$, and we define the *activity*

$$s_i = \text{clip}\left(\frac{1}{R} \text{curve}(d_i; e_i), 0, 1\right) \in [0, 1], \quad (1)$$

where R is a domain-specific normalizing scale ($R = 100$ for percent growth). The parameter vector itself is an information-rich summary of how the constituent will respond at any operating point, not just at d_i ; we z -score $e_{i,c}$ within a context to put every curve parameter on the same scale and write e_i^z for the result. The curve resolves how a constituent behaves in isolation; combination-specific effects must be modeled separately, which is the role of the CP-tensor structure.

3.3. From a naive curve embedding to the activity gate

The simplest way to put curve information into a kernel is to project the z -scored fingerprint as $f_i = M e_i^z$ through a learnable linear map $M \in \mathbb{R}^{q \times p}$ (this baseline is exactly the LINEAR_CURVE surrogate, which already accounts for a fraction of Savitar’s improvement). It is missing one piece of structure.

The vanishing constraint. A synergistic combination S should not contribute to the GP feature when any constituent i_j in S is operated at its no-effect point. Bliss-style independence formalizes this: an interaction term is informative only when all participating constituents are active. Let $g : [0, 1] \rightarrow [0, 1]$ be a smooth monotonic gate applied to the activity s_i and consider the interaction feature

$$\bigodot_{j \in S} f_{i_j} = \prod_{j \in S} g(s_{i_j}) \bigodot_{j \in S} M e_{i_j}^z. \quad (2)$$

For the right-hand side to vanish when any single constituent in S is inactive ($s_{i_j} = 1$), the factor $\prod_{j \in S} g(s_{i_j})$ must

equal zero, forcing $g(1) = 0$. Symmetrically, the interaction should be maximal when every constituent is fully active ($s_{i_j} = 0$), i.e. $g(0) = 1$. Among smooth monotonic functions on $[0, 1]$ satisfying these endpoints, $g(s) = 1 - s$ is the unique linear choice and the simplest non-trivial member of the family $g(s) = (1 - s)^\alpha$.

Activity-gated embedding. We adopt $g(s) = 1 - s$ throughout:

$$f_i = (1 - s_i) \cdot M e_i^z \in \mathbb{R}^q. \quad (3)$$

The factor $(1 - s_i)$ has a direct domain reading in each application (the per-drug kill fraction in pharmacology, the fractional saturation in chemistry, the moneyiness-adjusted leg activation in options), and it makes the no-effect-vanishing property automatic for all higher-order interactions.

3.4. Symmetric CP sharing

The most general parameterization of a symmetric order- m interaction over D constituents is a symmetric tensor $T^{(m)}$ with $\binom{D+m-1}{m}$ free parameters: for $D = 100$ and $m = 3$ that is 1.7×10^5 , more than the entire ALMANAC pool for any single cell line, with at most $T \leq 30$ observations to fit it. A rank- q symmetric CP decomposition instead gives

$$T_{i_1 \dots i_m}^{(m)} = \sum_{r=1}^q \prod_{j=1}^m A[i_j, r], \quad A \in \mathbb{R}^{D \times q}, \quad (4)$$

introducing Dq parameters regardless of $m_{\max}$ and arity k . For $D = 100$, $q = 8$ this is 800 parameters, comfortably below budget. Crucially, the same row $A[i, :]$ is reused for every order, which is the mechanism by which an observation at any combination involving i propagates to every other combination involving i .

3.5. Aggregation, kernel, and BO loop

The Hadamard product $\odot_{j \in S} f_{i_j} \in \mathbb{R}^q$ preserves both multiplication and dimension. Combining Equations (3) and (4) gives

$$\phi(c) = \sum_{j=1}^k f_{i_j} + \sum_{m=2}^{m_{\max}} \sum_{\substack{S \subseteq [k] \\ |S|=m}} w_S \cdot \odot_{j \in S} f_{i_j} \in \mathbb{R}^q, \quad (5)$$

with $w_S = \sum_r \prod_{j \in S} A[i_j, r]$. The activity gate factors out of every order- m term as $\prod_{j \in S} (1 - s_{i_j})$, so a synergistic interaction contributes feature mass only when every constituent in S is individually active. The final kernel is a radial basis function on ϕ :

$$k(c, c') = \sigma^2 \exp\left(-\frac{1}{2\ell^2} \|\phi(c) - \phi(c')\|^2\right), \quad (6)$$

with learnable signal variance σ^2 and lengthscale ℓ . Since ϕ is deterministic given (M, A) , k is the Gaussian RBF composed with a feature map and is therefore PSD on $\mathcal{C}$.

Cross-combination generalization follows: an observation at any combination containing constituent i propagates to the posterior at every other combination containing i , including unseen pairs and triplets, because $\partial \phi(c) / \partial A[i, :] \neq 0$ generically. Appendix B states and proves this.

BO loop. At step t we refit $\theta = \{M, A, \ell, \sigma^2, \sigma_n^2\}$ on the observations by Type-II marginal-likelihood maximization (30 Adam steps at lr 0.05), score each unobserved c by Expected Improvement, and query the argmax. The dominant cost is the Adam fit at ~ 0.03 s per step on CPU; end-to-end this yields a ~ 0.9 s BO trajectory on ALMANAC, 2–10 $\times$ faster than published GP baselines.

4. Cross-Domain Evaluation and Regime Boundaries

Table 1 gives the full cross-domain result. Savitar is evaluated on three domains under one fixed configuration ($q = 8$, $m_{\max} = 2$, ZeroMean, 30 Adam steps at lr 0.05) with no per-domain retuning, against random search, identity kernels (Hamming, activity-Hamming), each field’s published domain kernel (Tanimoto, GIP, INDIGO chemogenomic), flat and curve-aware Matérn GPs, deep ensembles, DeepSets, SAAS, and its own two structural ablations. In *drug discovery* it wins clearly where one drug’s effect depends on what another is doing (Brochado, an order of magnitude below the curve-aware baselines), ties the linear baseline where the dose–response signal is already strong (Cokol), improves 1.8 $\times$ on the chemistry-aware Tanimoto kernel on the legacy NCI-60 sweep, and is overtaken on Tan/HIV where the curve alone explains the data. In *chemistry* (Ni–Sn–Zn phase equilibria) SAAS narrowly wins the one phase with the fewest participating elements while Savitar beats every method on the two with more. In *finance* (same-day ETF options) its lead over the curve-only baseline widens as profitable strategies get rarer (0.0476 vs. 0.0703, $p = 0.0019$).

What carries the win, and where it stops. The two ablations are diagnostic and complementary: fixing $M = I_4$ ties full Savitar on Cokol but degrades badly on Brochado, while removing the activity gate is catastrophic on Cokol and mild on Brochado, so each contribution carries the win where the other is weakest. A capacity sweep finds $q \in \{4, 8\}$ optimal everywhere, so the benefit is the presence of structured interaction sharing rather than its capacity. The cells where Savitar is *not* best each violate one of the four regime properties: MOLT-4 has 6.7% of its pool within regret 5 of the optimum, so random search already finds it; on Tan/HIV and on the ALMANAC representative subset the per-constituent signal is dense enough that the linear curve baseline wins; on the smallest alloy phase the sparse axis-aligned prior wins. These are not failure modes but predic-

Table 1. Cross-domain final regret (mean over seeds) for every method in the Part I evaluation, reproduced from the predecessor paper (Gonehal, 2026). Bold is best per column; en dashes mark domain-inapplicable cells. Under a single fixed hyperparameter configuration, Savitar attains the lowest mean regret on eight of the ten primary benchmarks, losing Tan/HIV to `linear_curve` and `Ni3Zn14` to `saas`. Two patterns matter for Part II. First, the margin is largest exactly where the pool is large and winners are rare (ALMANAC, 5.54 against 9.16 for the strongest curve baseline and 70.4 for random), and it narrows or inverts on the small chemistry pools where a sparsity prior has less to overcome; this is the regime characterization Part II inherits when it restricts to in-regime cell lines. Second, the ablations locate the gain: removing the activity gate (`savitar_no_gate`) costs an order of magnitude on ALMANAC (82.2) and on Cokol (0.838 against 0.018), whereas removing higher-order interaction terms (`savitar_no_m`) is nearly free on the small pools and even wins three columns, so the gate carries the combinatorial-scale win and the tensor carries the interaction structure. Every number here is scored on well-mixed in vitro regret, the single scalar whose sufficiency Part II tests.

Method	Drug discovery				Chemistry (Ni–Sn–Zn)			Finance (ETF options)		
	ALMANAC	Cokol	Tan/HIV	Brochado	Ni ₃ Zn ₁₄	Sn	Ni ₃ Sn-HT	ETF fam.	SPY mo.	QQQ mo.
<i>Random and identity-only baselines</i>										
random	70.4	0.51	9.32	6.18	.019	.008	.020	.560	.373	1.12
hamming	56.2	–	–	–	–	–	–	–	–	–
act_hamming	56.2	–	–	–	–	–	–	–	–	–
<i>Published domain-aware baselines</i>										
tanimoto	79.1	–	13.45	–	–	–	–	–	–	–
gip	–	0.73	–	–	–	–	–	–	–	–
chemogenomic	–	–	–	3.30	–	–	–	–	–	–
<i>Flat / curve-aware GP baselines</i>										
flat_matern_id	81.8	–	–	–	–	–	–	–	–	–
flat_matern_curve	90.6	1.65	12.18	3.60	.024	.006	.018	–	–	–
linear_curve	9.16	0.020	2.65	1.25	.019	.0024	.0038	–	–	–
sphere_linear_curve	32.7	–	–	–	–	–	–	–	–	–
linear_descriptor	–	–	–	–	–	–	–	.339	.285	.502
linear_option_curve	–	–	–	–	–	–	–	.070	.075	.058
<i>Learned and set-function baselines</i>										
deep_ensemble	15.2	0.084	7.41	2.39	.019	.0029	.0044	.147	.110	.261
deepsets	13.3	0.247	8.16	7.76	.016	.008	.020	–	–	–
saas	–	–	–	–	.013	.0041	.009	–	–	–
<i>Savitar structural ablations</i>										
savitar_no_m	6.28	.014	13.86	1.15	.021	.0011	.024	.045	.060	.000
savitar_no_gate	82.2	0.838	14.52	0.57	.026	.0004	.013	.046	.061	.000
<i>Savitar (full)</i>										
savitar	5.54	.018	5.43	.040	.017	.0017	.0033	.048	.064	.000

tions of when the inductive bias should and should not help, and Part II takes that regime-dependence seriously, deriving its analysis cohort from the near-optimum-density criterion rather than assuming uniform applicability.

5. Justifying Rescoring

Every number in Part I is a scalar measured in a well-mixed monolayer: percent growth inhibition at a fixed exposure and time. That number is what the optimizer maximizes and what regret is computed from. A solid tumor is not well mixed: drug arrives by diffusion, oxygen falls off over $\sim 150\text{--}200\ \mu\text{m}$, and cells in the resulting hypoxic interior divide slowly and resist cycle-dependent agents. A combination potent in a monolayer may or may not clear that interior.

The predecessor work (Gonehal, 2026) anticipated the need for a pharmacological bridge and supplied one: Gompertz growth with linear PK, converting the BO objective into a simulated tumor burden. That bridge is well-mixed by construction, and we show in Section 7.1 that it is *exactly* rank-preserving: it cannot change which combination looks better, only restate the in vitro number on another scale. To

ask whether spatial structure changes the answer, the model must have space in it.

6. Spatial Model and Readouts

6.1. Analysis cohort and the out-of-regime control

Part I documents where curve-aware selection does and does not help, and the governing quantity is *near-optimum density*: the fraction of the pool within a small regret of the optimum. Where that fraction is large, random search already succeeds and no prior can add much. We therefore derive the cohort from the data rather than copying a list: recomputing near-optimum density for all 60 ALMANAC cell lines reproduces the two values the predecessor paper (Gonehal, 2026) reports (OVCAR-8 at 0.00046 against a published 0.05%; MOLT-4 at 0.0676 against 6.7%), and a threshold of 0.005 places 50 of 60 cell lines in regime. Applied blind to the eight cell lines of the predecessor’s ALMANAC study, it admits exactly seven and excludes exactly MOLT-4, the line that paper itself names as out of regime.

The seven in-regime lines (DU-145, HT29, OVCAR-8, MDA-MB-231/ATCC, MDA-MB-435, PC-3, T-47D) form the analysis cohort; MOLT-4 is retained as a labelled out-

of-regime control (Section 7.5). The split is not cosmetic: curve-aware selection beats the structural baseline on 45/50 in-regime lines versus 5/10 out of regime (Fisher $p = 0.0078$, OR = 9.0).

6.2. Spheroid reaction–diffusion model

We model a radially symmetric spheroid of radius $R = 400 \mu\text{m}$. The state is viable cell density $n(r, t)$, oxygen $c_O(r, t)$, and one field $c_j(r, t)$ per drug in EC50 multiples:

$$\partial_t n = D_n \nabla^2 n + \rho n(1 - n) \frac{c_O}{K_O + c_O} - \delta_{\text{hyp}} n \mathbf{1}[c_O < \theta] - K(c, c_O) n \quad (7)$$

$$0 = D_O \nabla^2 c_O - q_O n \frac{c_O}{K_O + c_O} \quad (8)$$

$$\partial_t c_j = D_j \nabla^2 c_j - u_j n \frac{c_j}{K_u + c_j} - \lambda_j c_j \quad (9)$$

Each drug contributes a Hill-form kill rate from its fitted monotherapy curve; drugs combine additively (Bliss-equivalent in the exponential-kill regime). Kill is modulated by cell-cycle dependence φ_j through the local proliferative state, so cycle-dependent agents lose potency in the hypoxic, quiescent interior: the mechanism that makes spatial structure matter. The spherical Laplacian is discretised on a uniform radial grid with the l’Hôpital limit $3f_{rr}$ at the origin; method of lines with LSODA on a node-interleaved (banded) Jacobian. The solver is verified against the analytic oxygen steady state and the Fisher–KPP front speed, and the oxygen consumption rate is calibrated so the viable rim falls in the 150–200 μm range characteristic of spheroids ($q_O = 15/\text{day}$, giving a 166 μm rim, hypoxic fraction 0.203, necrotic fraction 0.058 in the untreated control; none of which were fitted). Appendix C gives the error figures.

6.3. Readouts anchored to patient outcome

We surveyed the tumour-growth-dynamics literature (58 primary papers, 55 DOI-verified) for quantities extractable from a simulated burden trajectory that have published patient-level outcome correlations, grading each by evidence tier (A: validated against patient survival; B: animal/organoid; C: mechanistic only).

The most consequential concerns the two-phase model $N(t)/N(0) = e^{-K_S t} + e^{K_G t} - 1$. In 112 metastatic castration-resistant prostate cancer patients, K_G predicts overall survival ($r = -0.72$) while K_S does not ($r = -0.218$); above-median K_G carries a mortality hazard ratio of 5.14 (95% CI 3.10–8.52). This matters because K_S is the quantity a monolayer assay measures, and K_G is not. An optimiser scored on in vitro potency is optimising the constant that does not predict survival. We report evidence tiers alongside every readout and do not claim a simulated K_G is a patient K_G (Section 9).

6.4. Simulation protocol

Every distinct combination selected by any method was simulated with a matched untreated control in the same cell line, 21-day horizon, five daily doses. The primary readout is log-kill AUC, the time-integrated burden reduction relative to control, which we adopt because end-of-horizon log-kill saturates (all arms relax to the same attractor) and is approximately $40\times$ less discriminating.

7. Spatial Results

7.1. The published ODE bridge is exactly rank-preserving

We first reproduce the predecessor paper’s (Gonehal, 2026) well-mixed bridge as a control. All six published table rows reproduce to a maximum absolute error of 0.0139 log units (mean 0.0051), including the no-treatment row. Over the 1,397 unique combinations, the rank correlation between in vitro percent growth and simulated log-reduction is $\rho = -1.0000$ ($p = 0$): the bridge is a monotone re-expression of the in vitro number. Any re-ordering we observe below is therefore attributable to spatial structure, not to the act of simulating.

7.2. Spatial simulation re-orders combinations

In the spheroid, the same relationship is only moderate (Figure 3a): $\rho = -0.56$ for log-kill AUC against percent growth, and 36.5% of within-cell-line combination pairs are re-ordered relative to their in vitro ranking. The gap is not noise but resolution: among 98 combinations spanning less than 10 percentage points of in vitro potency, spheroid log-kill AUC spans 1.30 log units (Figure 1). In vitro potency explains 23% of the variance in spheroid outcome; adding a single spatial quantity (residual hypoxic fraction) raises this to 86%.

Separately, the readout with the strongest published survival association, K_G , is statistically unrelated to in vitro potency ($\rho = 0.004$, $p = 0.95$).

7.3. Method comparison under a spatial model

Table 2 ranks all sixteen methods by simulated K_G (the readout carrying the strongest published survival association) rather than by the spatial readout; the two orderings agree closely (Spearman $\rho = -0.86$) and Savitar places fifth under either.

Savitar beats every structural baseline by a wide, highly significant margin: Tanimoto (+0.299 median log-kill AUC, Holm $p < 0.001$), Hamming and activity-Hamming (+0.230), random (+0.158), both flat-Matérn variants, and both of its own ablations. This reproduces the predeces-

Table 2. Sixteen selection methods on the seven in-regime cell lines, 10 seeds each, ranked by simulated K_G : the tumor regrowth-rate constant, which among our readouts carries the strongest published association with patient overall survival (Section 6.3). Lower K_G is better. p is a paired Wilcoxon test against the curve-aware method on log-kill AUC, Holm-corrected over the 15 comparisons. Ranking by K_G and by log-kill AUC agree closely (Spearman $\rho = -0.86$).

Method	Outcome-anchored		Spatial		In vitro		p
	$K_G \downarrow$	TTG	Log-kill	Hypoxic	% growth	Regret	
DeepSets	0.0229	2.66	0.624	0.091	-71.7	22.1	0.008
SAASBO	0.0248	2.76	0.582	0.124	-86.4	7.4	0.126
Linear (curve)	0.0254	2.94	0.552	0.123	-81.7	12.1	0.243
Deep ensemble	0.0256	2.94	0.533	0.130	-76.8	17.0	0.382
Curve-aware (ours)	0.0290	3.19	0.453	0.147	-79.2	14.6	–
Spherical linear	0.0291	4.53	0.311	0.163	-52.2	41.5	0.008
Matérn (curve)	0.0295	5.42	0.173	0.191	-15.8	78.0	<0.001
Matérn (identity)	0.0297	4.99	0.248	0.178	-16.5	77.3	<0.001
Ours: no M (ablation)	0.0297	3.19	0.438	0.151	-71.0	22.8	0.822
Random	0.0302	4.66	0.226	0.183	-23.4	70.4	<0.001
COMBO (graph diffusion)	0.0308	5.06	0.182	0.195	-2.3	91.5	<0.001
Tanimoto	0.0310	5.86	0.140	0.197	-20.5	73.3	<0.001
BOCS (sparse Bayes)	0.0318	5.42	0.199	0.189	-14.9	78.9	<0.001
Ours: no gate (ablation)	0.0354	4.45	0.233	0.191	-19.2	74.6	<0.001
Activity-Hamming	0.0359	5.57	0.165	0.197	-28.4	65.3	<0.001
Hamming	0.0359	5.57	0.165	0.197	-28.4	65.3	<0.001

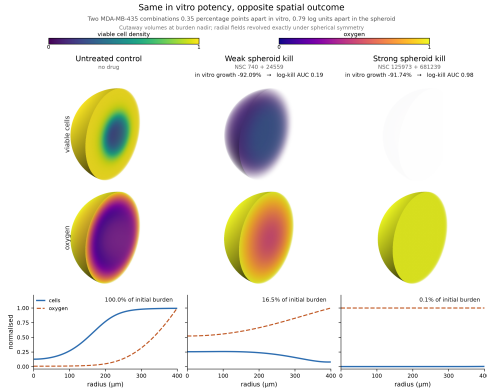


Figure 1. Volume renderings of the simulated spheroid at burden nadir for two combinations in MDA-MB-435 that are matched in vitro (-92.09 vs -91.74 percent growth, 0.35 points apart) but separated by 0.79 log units in the spheroid. Left: viable cell density; right: oxygen. The upper combination leaves 16.5% of the initial burden behind a spared hypoxic core; the lower clears to 0.1%. Holding assayed potency fixed isolates the spatial mechanism from drug strength. Rendering integrates the emission-absorption volume integral over the revolved radial solution, exact for a spherically symmetric field.

sor (Gonehal, 2026) paper’s finding under a scoring model it was never tuned for. Four higher-capacity surrogates rank above it, but after Holm correction across fifteen comparisons only DeepSets does so significantly (-0.045 median, $p = 0.008$); SAASBO ($p = 0.126$), the linear curve baseline ($p = 0.243$) and deep ensembles ($p = 0.382$) are statistically indistinguishable. On the spatial readout Savitar

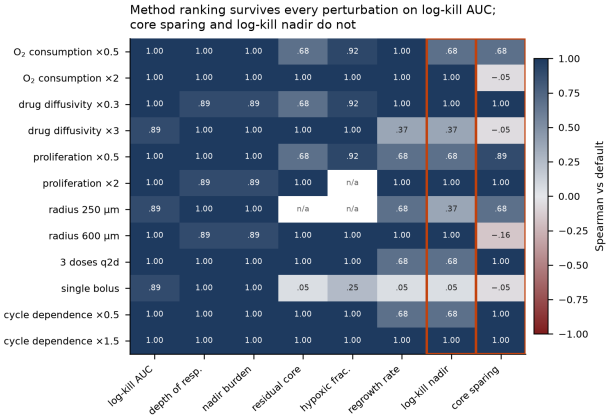


Figure 2. Robustness of each readout’s method ordering across thirteen parameter perturbations (Spearman against default). Median $\rho = 1.000$; $\rho \geq 0.8$ in 74.2% of computable cells. Log-kill AUC, which carries the comparison, never falls below 0.895. Core sparing and log-kill at nadir (outlined) are unstable and reported as diagnostics only.

places fifth of sixteen (beating ten methods significantly, tied with four, beaten by one).

The two leaders lead for different reasons. SAASBO also attains the best in vitro regret of the sixteen (7.4) and the strongest in vitro potency (-86.4% growth): its spatial standing follows from solving the original search problem better, at roughly $600\times$ Savitar’s cost (182 s versus 0.3 s per BO run at this pool size). DeepSets does not: it ranks fifth

on regret, below Savitar, while placing first spatially. Section 7.4 separates these cases quantitatively.

A new comparison the predecessor’s regime account does not predict. That paper evaluates SAASBO only on its chemistry benchmark, never on the drug panel, so this ALMANAC comparison is run for the first time. Its account is explicit: SAAS sparsifies which input *dimensions* matter, Savitar parameterizes which *combinations* interact, so SAAS wins only on the alloy phase with a small constituent universe. ALMANAC’s universe is large (~ 95 drugs), so that account predicts Savitar should be favored. It is not: on regret SAASBO wins clearly (7.40 vs. 14.58, better in 51 of 70 runs and in all seven cell lines, $p < 10^{-4}$), while on the spatial readout the two are tied. Which prior wins does not reduce to constituent-universe size alone.

Dedicated combinatorial-BO methods do not transfer to this pool. We additionally run COMBO (Oh et al., 2019), a GP with a diffusion kernel on the graph Cartesian product of drug identity and ordinal dose, and BOCS, sparse Bayesian regression over first- and second-order binary indicators, under the same pool, budget and initialization, with a deviation *favoring* BOCS, enumerating the finite pool exactly at the acquisition step so it never loses to an inner-optimizer failure. On three anti-infective screens with pools of ~ 64 –100 candidates both are competitive and curve-aware selection beats them in all six comparisons (margins 0.08–4.42 regret units, significant in five of six). On ALMANAC, whose enumerated pool is roughly $700 \times$ larger ($\sim 4.7 \times 10^4$ combinations per cell line), both collapse: mean regret 91.5 (COMBO) and 78.9 (BOCS) against 14.6, worse than random search (70.4), and both are beaten significantly on log-kill AUC ($+0.188$ and $+0.266$ median, Holm $p < 0.001$) and on K_G . Neither consumes the response curve (COMBO sees dose only as an ordinal level on a path graph, BOCS as a scalar), so at 30 evaluations against the 4,000-candidate subsample every method scores at each acquisition step (Appendix E), neither can generalize across unseen pairs. At small pool sizes a generic combinatorial prior suffices; at this one it does not.

7.4. Mechanism: potency without penetration

The deficit is mechanistic and legible in the table. Across the sixteen methods, spheroid performance tracks *residual hypoxic fraction* at $\rho = -0.969$: more tightly than it tracks in vitro potency ($\rho = -0.767$). Hypoxic fraction is purely spatial: not measurable in a monolayer, and absent from every method’s objective.

To separate “selects better combinations” from “searches better,” we regress each method’s spatial performance on its in vitro potency and read the residual: spheroid kill obtained beyond what assayed potency predicts. This cleanly

separates the two leaders. DeepSets is the largest positive outlier ($+0.140$): it selects combinations that penetrate and leaves the least hypoxic tissue of any method (0.091). SAASBO sits on the trend line ($+0.014$): it leads spatially because it is the strongest in vitro searcher, not because it selects better-transporting combinations. Savitar is the largest negative outlier (-0.074): its picks are *more* potent in vitro than DeepSets’ (-79.2 vs. -71.7 percent growth) yet leave 62% more hypoxic tissue (0.147 vs. 0.091). It converts in vitro potency into spheroid kill less efficiently than any method in the comparison.

This follows from its design rather than indicating a defect in it. The curve-aware kernel identifies pairs whose dose–response curves interact favorably *at the assayed exposure*, which is why it wins on regret. Nothing in that objective rewards a drug reaching an oxygen-poor core, or penalizes one whose partner is cycle-dependent and therefore inert in quiescent tissue. The two tasks overlap heavily but are not the same task.

A shared ceiling. All sixteen methods are scored on the same monolayer curves, so this input is constant across the comparison and cannot itself explain the re-ranking. It does bound what any of them can achieve: an in vitro curve carries no transport information, and a method optimizing it perfectly would still be blind to penetration. The re-ranking is caused by how efficiently each method converts that limited signal into spatial performance.

7.5. The out-of-regime cell line

MOLT-4 is reported separately (Table 4) because it fails the regime criterion of Section 6.1: 6.7% of its pool lies within a small regret of the optimum, against 0.05% for OVCAR-8. The consequence is visible in the table: every method drives percent growth to ≈ -99 and regrets compress to under 2, so the in vitro task carries almost no signal to discriminate on. Four methods tie exactly at the pool optimum. The spatial readouts still separate them, and the rank agreement between in vitro and spatial orderings collapses (Kendall $\tau = 0.13$ against 0.52 in regime). Pooling MOLT-4 into the main comparison flips three methods from tied to significantly better than Savitar (SAASBO, the linear curve baseline and deep ensembles), which is why it is kept separate and labeled rather than dropped or merged.

7.6. Robustness

A conclusion that holds only at default settings is not a conclusion. Across thirteen parameter settings perturbing oxygen consumption, diffusivity, proliferation, radius, schedule and cycle-dependence, the method ordering is stable in the median (Spearman $\rho = 1.000$ against default; $\rho \geq 0.8$ in 69 of 93 computable cells), the curve-aware method ranks

first on log-kill AUC in *all thirteen* settings, and log-kill AUC never falls below $\rho = 0.895$. Core sparing inverts under four settings and log kill at nadir has median $\rho = 0.684$, so we report the method comparison as robust and those two orderings as not established (Appendix G, Figure 2).

8. Discussion and Future Work

The natural reading of these results is not that curve-aware BO is misconceived, but that it is currently optimizing an objective that stops short of the phenomenon of interest. Saviar wins decisively on the target it was given, and the target omits transport. What SAAS supplies is a half-Cauchy prior over per-dimension inverse lengthscales; the curve-aware kernel uses a single isotropic ℓ over $\phi(c)$, treating every coordinate as a priori equally relevant. The concrete extension is a diagonal ARD lengthscale under a sparsity-inducing hyperprior, fit by the Type-II step already in the loop. It is orthogonal to both structural priors (the gate and CP sharing govern how ϕ is *built*, the lengthscale prior how it is *compared*), and it targets Section 7.4 directly: with transport descriptors appended to the fingerprint, ARD is the mechanism that would let the surrogate discover a transport coordinate matters while most curve coordinates do not. A MAP fit retains the bias without SAASBO's ~ 182 s NUTS inference. Appendix A develops this and two further routes.

9. Limitations

The model is calibrated, not validated: it reproduces spheroid layering and passes analytic benchmarks (Appendix C), but no spheroid was grown and no combination tested, so the claim is comparative (that spatial structure reorders selections), not that any absolute number predicts an experiment. Drug transport is assigned by mechanism class rather than measured; Appendix G bounds the consequence. Readout anchoring is not readout validation: a simulated K_G is a different quantity from a patient K_G , sharing a name and a functional form. The geometry is radial and avascular: no vasculature, interstitial pressure, immune compartment or resistance evolution. Scope is seven in-regime cell lines plus one out-of-regime control from a single panel, at one horizon and schedule.

10. Conclusion

A well-mixed bridge from in vitro potency to tumor burden is exactly rank-preserving and cannot change which combination looks better (Appendix F); a spatially resolved spheroid can, and does, for 36.5% of pairs. Under that model curve-aware selection keeps a large advantage over every structural baseline and over two dedicated combinatorial-BO methods, but falls behind higher-capacity surrogates selecting better-penetrating combina-

tions. The cause is identifiable: outcome is governed by residual hypoxic fraction ($\rho = -0.969$), absent from every method's objective, and points at a fix: a sparsity-aware lengthscale prior over an interaction-structured feature.

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Table 3. Curve-aware selection versus SAASBO on every cell line of the predecessor paper’s (Gonehal, 2026) ALMANAC set, 10 seeds each. SAASBO attains lower regret in *all eight* (significantly in three), and higher spheroid log-kill AUC in six of eight. p is a paired Wilcoxon test, uncorrected. [†]MOLT-4 lies outside the published operating regime.

Cell line	Regret (in vitro)			Log-kill AUC (spatial)		
	Ours	SAAS	p	Ours	SAAS	p
MDA-MB-231/ATCC	13.39	0.01	0.031	0.574	0.938	0.031
OVCAR-8	5.52	0.25	0.219	0.505	0.260	0.125
MDA-MB-435	2.67	1.30	0.107	0.270	0.421	0.188
T-47D	5.58	1.34	0.002	0.430	0.365	0.205
PC-3	14.35	8.16	0.074	0.358	0.422	0.504
HT29	21.37	17.43	0.797	0.467	0.572	0.156
DU-145	39.19	23.29	0.002	0.568	1.098	0.049
MOLT-4 [†]	1.09	0.41	0.129	0.458	1.566	0.004

Table 4. MOLT-4, the **out-of-regime** control, same 16 methods and ranking. It is excluded from Table 2 because 6.7% of its pool lies within a small regret of the optimum (versus 0.05% for OVCAR-8), so random search already succeeds, every method drives percent growth to ≈ -99 , and the in vitro task carries almost no signal to discriminate on. The spatial readouts still separate the methods, and the in vitro-to-spatial rank agreement collapses here (Kendall $\tau = 0.13$ versus 0.52 in regime).

Method	$K_G \downarrow$	Log-kill	Hypoxic	% growth	Regret
SAASBO	0.0052	1.566	0.004	-99.5	0.41
Linear (curve)	0.0068	1.382	0.000	-99.5	0.39
Deep ensemble	0.0124	1.259	0.045	-99.4	0.51
DeepSets	0.0156	0.755	0.078	-98.2	1.76
Spherical linear	0.0160	1.022	0.061	-99.5	0.40
COMBO (graph diffusion)	0.0190	0.904	0.097	-93.9	6.00
Matérn (identity)	0.0191	0.930	0.085	-97.8	2.09
BOCS (sparse Bayes)	0.0196	0.823	0.077	-95.2	4.74
Matérn (curve)	0.0217	0.702	0.081	-97.5	2.41
Random	0.0286	0.504	0.132	-96.7	3.22
Curve-aware (ours)	0.0310	0.458	0.160	-98.8	1.09
Activity-Hamming	0.0314	0.447	0.178	-99.2	0.73
Hamming	0.0314	0.447	0.178	-99.2	0.73
Ours: no M (ablation)	0.0314	0.447	0.178	-99.2	0.73
Tanimoto	0.0314	0.447	0.178	-99.2	0.73
Ours: no gate (ablation)	0.0315	0.426	0.174	-99.2	0.66

A. Extended Discussion: Toward a Spatially Aware Kernel

The sparsity prior. What SAAS supplies that the curve-aware kernel does not is a half-Cauchy prior over per-dimension inverse lengthscales, letting the surrogate infer that most input coordinates are irrelevant and concentrate limited evidence on the few that are not. The curve-aware RBF (Equation (6)) uses a single isotropic ℓ over $\phi(c)$: every coordinate is a priori equally relevant. At a 30-evaluation budget this is a real cost, since the curve fingerprint contributes p parameters per constituent of which typically one or two carry the signal for a given cell line. The extension is to replace the isotropic ℓ with a diagonal ARD lengthscale $\ell \in \mathbb{R}^q$ under a sparsity-inducing half-Cauchy hyperprior, fit by the Type-II marginal-likelihood step already in the loop. This is orthogonal to both structural priors (the activity gate and CP sharing act on how ϕ is *built*, the lengthscale prior on how ϕ is *compared*), so the two should compose rather than compete. What makes SAASBO impractical at this pool size is its fully Bayesian NUTS inference (~ 182 s per BO run against ~ 0.3 s); an ARD lengthscale with a MAP fit retains the inductive bias at a small fraction of that cost.

Three routes. *Objective augmentation:* score candidates on a cheap surrogate of penetration (predicted tissue diffusivity and cycle-dependence from mechanism of action) alongside curve interaction, so that two combinations of equal assayed potency are no longer equivalent to the optimizer. *Fingerprint extension:* the per-constituent fingerprint e_i is already a small parameter vector fit upstream, so appending transport descriptors requires no change to the kernel algebra; combined with the ARD prior above it would let the model learn that steep-curve pairs which are both cycle-dependent are a poor bet

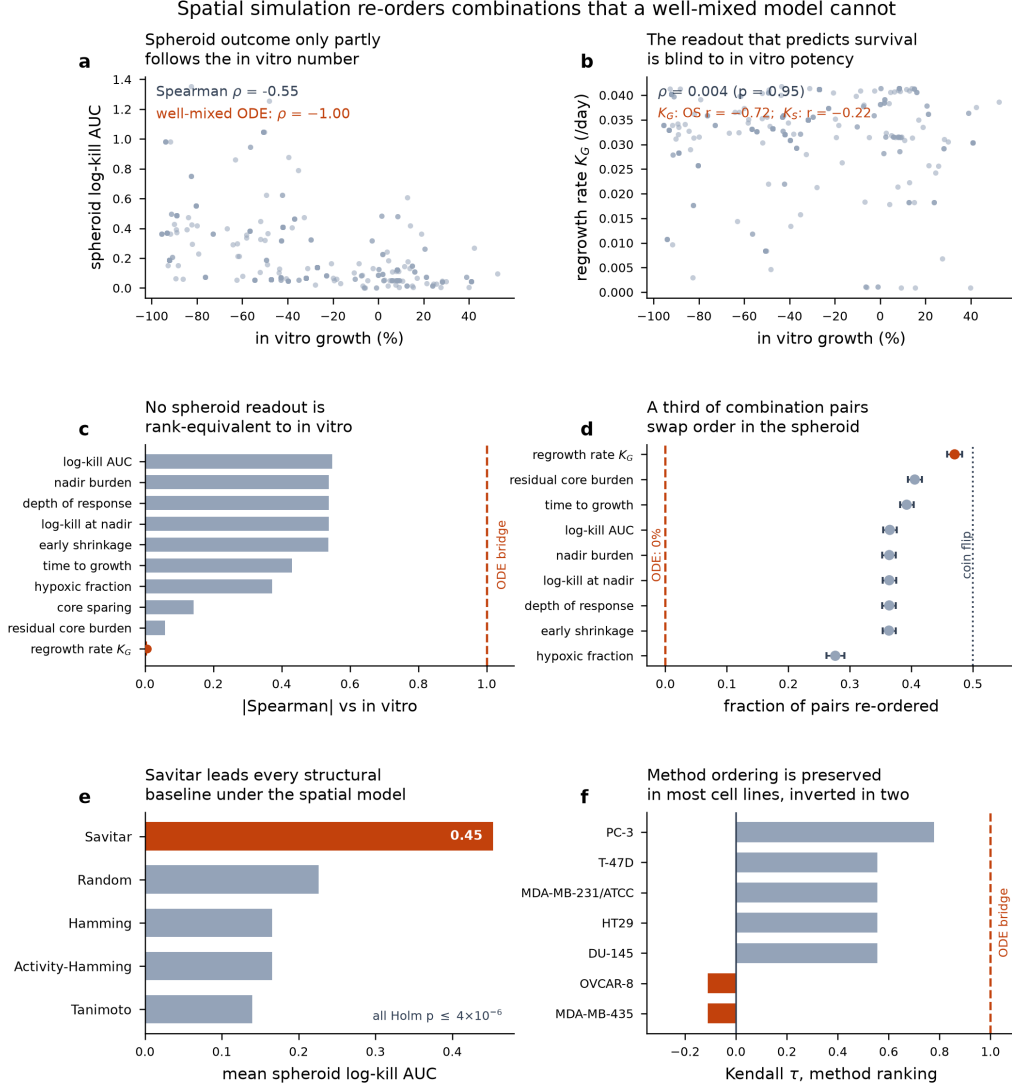


Figure 3. Spatial simulation re-orders combinations that a well-mixed model cannot. (a) Spheroid log-kill AUC against the in vitro potency that selected each combination ($\rho = -0.56$); the well-mixed ODE bridge would place every point on a monotone curve ($\rho = -1.00$). (b) The clinically anchored regrowth rate K_G is unrelated to in vitro potency ($\rho = 0.004$, $p = 0.95$), while K_G (not the kill constant K_S) is the quantity that predicts overall survival in patients. (c) Monotonicity of each readout against in vitro potency. (d) Pairwise re-ranking rate with bootstrap CIs. (e) Mean log-kill AUC by method. (f) Per-cell-line rank agreement.

in hypoxic tissue. *Simulation in the loop*: the solver runs ~ 1 s per combination, fast enough to serve as the BO objective directly at the ~ 30 -evaluation budgets these campaigns use, turning the spheroid from an evaluation instrument into the thing being optimized. The comparison also supplies a target: the spatial deficit is quantified as a residual against the potency trend, so an extended method can be scored on whether it closes the 0.21 log-unit gap to DeepSets while retaining its regret advantage.

B. Cross-Combination Generalization

Theorem 1 (Cross-combination generalization). *For any combination $c \in \mathcal{C}$ and any constituent i appearing in c , the partial derivative $\partial\phi(c)/\partial A[i, \cdot]$ is non-zero whenever $|S| \geq 2$ for some $S \subseteq [k]$ containing slot j with $i_j = i$. Consequently, an observation at any combination c' containing constituent i propagates information to the posterior predictions at every other combination c containing i , including unseen pairs and triplets.*

Proof. Differentiating Equation (4) yields $\partial w_S / \partial A[i, r] = \prod_{j' \in S \setminus \{j\}} A[i_{j'}, r]$ whenever some $j \in S$ has $i_j = i$, which is non-zero generically in A . The Gaussian-RBF kernel of Equation (6) is a smooth function of ϕ , so non-zero gradients of ϕ propagate to non-zero gradients of $k(c, c')$ at any pair sharing constituent i . $\square$

C. Solver Verification

The radial reaction–diffusion solver is verified against two closed-form benchmarks before any study result is computed.

Oxygen steady state. With proliferation disabled, the oxygen field must relax to the analytic steady state of the spherical diffusion–consumption equation. Measured relative L_2 error: 5.47×10^{-3} on the production mesh.

Fisher–KPP front speed. With diffusion and logistic growth only, a travelling front propagates at $c = 2\sqrt{D\rho}$. Seeding a plug at $r < 6000 \mu\text{m}$ in a radius-12000 μm domain and fitting only $t > 20$ days (to exclude the transient) gives 169.5 versus the analytic 178.9, a 5.2% error at a mesh ratio of 3.37. An earlier measurement taken near the origin during the transient gave a spurious 42% error; the benchmark must be measured in the travelling-wave regime.

Jacobian banding. The state is interleaved by node ($[n, O_2, c_1, c_2]$ at r_0 , then at $r_1, \dots$) rather than by field, which makes the Jacobian banded with half-bandwidth $2n_f - 1$. Supplying `lband/uband` to the stiff integrator reduces a two-drug simulation from 298.3 s to 2.99 s, a $100\times$ speedup with bitwise identical readouts.

D. Readout Construction

Why time-integrated exposure. Drug penetration cannot be judged from a concentration snapshot. With $\tau = R^2/D \approx 3.2$ days against a 0.25-day dose, a snapshot at peak surface concentration reports zero or undefined core penetration for every drug. Measuring the time-integrated exposure $\text{AUC}(r) = \int c(r, t) dt$ instead separates transport tiers cleanly: core-to-surface ratio 0.454 for a slow drug versus 0.862 for a fast one.

Why log-kill AUC rather than end-of-horizon log kill. By day 21 every arm has relaxed toward the same carrying-capacity attractor. The end-of-horizon log kill spreads only 0.0026 across all arms; the time-integrated log-kill AUC spreads 0.0896 and the log kill at nadir 0.1118 (roughly $40\times$ more discriminating). Regional kill fractions are referenced against the matched untreated control at the same time point rather than against $t = 0$.

Oxygen consumption is calibrated, not assumed. The consumption rate is set so the viable rim of a 400 μm spheroid falls inside the 150–200 μm oxygen-diffusion limit: $q = 14$ gives 180.9 μm and $q = 16$ gives 164.8 μm , so we use $q = 15$. With hypoxic death at 0.25/day the untreated 21-day spheroid shows a viable rim of 166 μm , hypoxic fraction 0.203 and necrotic fraction 0.058, reproducing the layered structure the model is meant to represent.

E. Reproduction of the Published Pipeline

Regenerating the ALMANAC combination table from the raw 615 MB source is bit-identical to the shipped artifact (2,804,088/2,804,088 rows, max absolute difference 0). Hill fits reproduce at Pearson $r = 0.998\text{--}0.9998$ across parameters, with 13/6,293 rows (0.21%) differing from optimizer basin flips under a newer SciPy. BO selections reproduce 138/144 (95.8%) bit-exactly. A dedicated re-run of three methods over the analysis cohort matched all 210/210 selections on drug identity, percent growth and regret, with concentration columns agreeing to float round-trip precision ($\leq 7.5 \times 10^{-17}$).

F. The Well-Mixed ODE Control

The predecessor paper’s (Gonehal, 2026) Gompertz + linear-PK bridge is reimplemented and validated against its published table. All six rows reproduce, including the no-treatment row, to a maximum absolute error of 0.0139 log units (mean 0.0051). Applied across 1,397 distinct combinations, the bridge’s rank correlation with in vitro percent growth is Spearman $\rho = -1.0000$ ($p = 0$): it is exactly rank-preserving and therefore cannot re-order any pair of combinations. This is the control against which the spheroid’s $\rho = -0.56$ should be read.

G. Sensitivity Sweep Detail

Thirteen parameter settings perturb oxygen consumption, drug diffusivity, proliferation rate, spheroid radius, dose schedule and cycle-dependence. Median Spearman correlation of the method ordering against the default setting is 1.000, with $\rho \geq 0.8$ in 69 of 93 computable cells (74.2%; three cells are undefined where a readout is constant). Per-readout medians: depth of response, log-kill AUC, nadir burden, residual core burden and hypoxic fraction all exactly 1.000; regrowth rate 0.842; core sparing 0.789; log kill at nadir 0.684. The bolus schedule is the most disruptive single perturbation, degrading five of eight readouts below $\rho = 0.3$. Core sparing inverts outright under four settings, which is why it is reported as a mechanistic diagnostic rather than a ranking quantity.

H. Repository Audit

An audit of the code released with the predecessor manuscript ([Gonehal, 2026](#)) found the pipeline reproducible from the shipped intermediates but incomplete at the edges: the raw NCI-ALMANAC source file and a chemical structure file required by one baseline are absent (every published NCI download URL now returns 403/404), the shipped ALMANAC results cover 8 cell lines $\times$ 3 seeds against the manuscript’s 60 $\times$ 10, one results file is misfiled under the wrong benchmark, and the BO traces record regret without recording the selected combination, so the selections analysed here had to be recovered by re-running the loop with the trace extended. None of these contradicts a published claim; together they mean the headline table cannot be regenerated from the repository alone.